

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2023
 Programme: BE Full Marks: 100
 Course: Engineering Economics Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why we need to study engineering economics? State and explain law of demand and supply. 2+5

- b) What do you mean by IRR? Evaluate IRR of the following project and decide whether the project acceptable or not? Where MARR 8% per year. 8

Initial investment =Rs. 5,000,000

Expected life =10 Years

Salvage value =Rs. 30,000

Annual revenue =Rs. 120,000

2. a) How do you measure following cost concept in your engineering project; direct cost, indirect cost, average cost, variable cost, marginal cost, sunk cost, opportunity cost? 8

- b) From the following project information select the best project using capitalized worth method. MARR = 10% 7

Items	X	Y	Z
Initial Investment	50,000	40,000	30,000
Annual Revenue	20,000	15,000	14,000
Annual Expenses	15,000	10,000	8,000
Useful life year	5	7	9
Salvage value	1000	500	0

3. a) What is the future equivalent of Rs. 500,000.00 per year uniform cash flows for 10 years if nominal rate is 10% compounded continuously? 8

- b) Discuss economic theory and ecological footprint. In the pursuit of achieving global sustainable goals, what vital role should engineers play to overcome ecological limits and promote environmentally responsible practices? 7

4. a) Find the B/C ratio by both methods. Use AW method. Investment = Rs.2,50,000; Annual benefits = Rs.75,000; Annual cost = Rs.15,000; salvage value = Rs.25,000; MARR = 12%; Number of years = 10. 8

- b) What do you mean by project financing and project funding? Explain with some example. 7

5. a) By using PW model & repeatability assumption select the best project. 7

Project	A	B
Initial Investment Rs	4,00,000	6,00,000
Annual Revenue Rs.	30,000	35,000
Annual O &M Rs.	3,000	4,000
Useful life in year	6	8
Salvage value Rs	4,000	7,000
MARR	12%	12%

- b) What are financial statements? Explain the types of financial statements with their formats. 8

6. a) Perform sensitivity analysis of the following project over the range of $\pm 20\%$ for 8

i. initial investment

ii. annual revenue

iii. useful life &

iv. MARR.

Initial investment = 11, 500

Annual revenue = 3,000

Salvage value =1,000

Useful life = 6 years

MARR = 12%

- b) A new electric sawmill for cutting small pieces of lumber in a furniture manufacturing plant has a cost basis of Rs. 5,000 and 8 year depreciable life. The estimated salvage value of the saw is zero at the end of 4th year cumulative depreciation charge of 7th year and book value of 6th year using double declining balance method. 7

7. Write short notes on: (**Any two**) 2×5

a) Importance of financial statement and accounting equation.

b) Financial Ratios

c) Ecology and economic development.